ModelIt!

Teacher's Guide

Why Your Phone Battery Dies So Fast

Energy Conversion, Efficiency, and the Heat Tax

Quick Reference

Grade Level	Grade 09L1
NGSS Standard	HS-PS3-1, HS-PS3-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-PS3-1, HS-PS3-3

Create a computational model to calculate the change in the energy of one component in a system when the change in energy of the other component(s) and energy flows in and out of the system are known. Design, build, and refine a device that works within given constraints to convert one form of energy into another form of energy.

Three-Dimensional Learning

SEP	**Using Mathematics and Computational Thinking** — Students create computational models to calculate energy changes and predict battery life under different usage scenarios.
DCI	**PS3.A: Definitions of Energy / PS3.D: Energy in Chemical Processes** — Energy is stored in batteries as chemical energy and converted through multiple steps. At each conversion, some energy is dissipated as thermal energy, reducing efficiency.
CCC	**Energy and Matter: Flows, Cycles, and Conservation** — Energy is conserved but degrades in quality — chemical energy converts to electrical to light/computation plus waste heat. Tracking energy flows reveals why systems are never perfectly efficient.

CAST Testing Alignment

CALCULATE:

A phone battery stores 15.4 Wh of energy. The screen uses 2.5 W, the processor uses 3.0 W during gaming, and 0.8 W of heat is generated. How long can the phone game before dying? What's the energy efficiency?

EXPLAIN:

Two identical phones with identical batteries are used for 4 hours — one for texting, one for gaming. The gaming phone is nearly dead while the texting phone has 70% battery. Use your energy model to explain why.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Every device you own is a chain of energy conversions, and every link in that chain leaks energy as heat. Understanding why requires grasping the fundamental physics of energy transformation and the inescapable reality of the second law of thermodynamics.

How Batteries Store Energy

Lithium-ion batteries store energy in chemical bonds. During discharge, lithium ions flow from the anode to the cathode through an electrolyte, releasing electrons that travel through the external circuit (your phone's components) as electrical current. A typical smartphone battery stores about 15 Watt-hours of energy — equivalent to lifting a 1-kg weight about 5,500 meters.

The Energy Conversion Chain

Battery chemical energy → electrical current → multiple endpoints: screen (electrical → light), processor (electrical → computation + heat), speakers (electrical → sound), radios (electrical → electromagnetic waves). At every conversion step, the second law of thermodynamics demands that some energy degrades to thermal energy. No conversion is 100% efficient.

Why Phones Get Hot

Heat generation in phones comes from electrical resistance in circuits (Joule heating), transistor switching in processors, and LED/OLED inefficiency in screens. A phone running a demanding game might generate 3-5 watts of heat — equivalent to a small nightlight. This is energy your battery paid for but that did nothing useful. Modern phones use thermal throttling to reduce performance when temperatures get dangerous.

Battery Degradation

Lithium-ion batteries degrade with every charge cycle. Repeated expansion and contraction of electrode materials causes microscopic cracks, and a thin layer of lithium compounds (SEI layer) gradually grows thicker, trapping lithium ions permanently. After 500 full cycles, most batteries retain only 80% of original capacity. Fast charging and extreme temperatures accelerate degradation.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify screen brightness, processing load, battery capacity, heat generation, and battery life as components of the phone energy system.

E **Establish Relationships**
Students discover that brightness and processing load both drain battery (+heat, -life) while heat represents wasted energy.

V **Visualize & Model**
Students build a model showing energy flowing from battery through components with heat as a loss pathway.

E **Evaluate Outcomes**
Students run gaming, standby, and brightness test scenarios to compare energy drain rates.

R **Revise & Extend**
Students add battery age, ambient temperature, or background apps to create a more realistic energy model.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Raise your hand if your phone has ever died at the worst possible moment. Why does this happen?"*

Watch For

Quick poll: What drains your battery fastest? Games? TikTok? YouTube?

Slide 2: Learning Objectives

3 min

Say: *"Today we're investigating the physics of why your phone battery can't keep up with your life."*

Watch For

Have students read objectives. Pre-teach 'energy conversion' with a simple example.

Slide 3: Big Question

3 min

Say: *"3 hours gaming vs 24 hours standby — same battery. Why the 8x difference?"*

Watch For

Show actual battery drain curves. Ask students to hypothesize.

Slide 4: LEVER Framework

2 min

Say: *"We'll model where your battery energy actually goes — and how much is wasted."*

Watch For

Preview LEVER steps. Emphasize we're tracking energy FLOWS.

Slide 5: Activity 1: Components

8 min

Say: *"What can YOU control about your phone's energy use? What happens as a RESULT?"*

Watch For

Guide sorting. Discuss why brightness is external (you choose) but heat is internal (you can't avoid it).

Slide 6: Activity 2: Connections

8 min

Say: *"More brightness means more battery drain. But where does that extra energy actually GO?"*

Watch For

Reveal that heat is the hidden energy thief. Draw the energy flow diagram.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's race: gaming mode vs standby vs brightness-only. Which kills your battery fastest?"*

Watch For

Students predict battery life for each scenario. Compare predictions to simulation.

Slide 8: Discoveries

5 min

Say: *"So every time your phone gets hot, that's YOUR battery energy being wasted as heat. The second law of thermodynamics strikes again."*

Watch For

Connect heat to wasted energy. Discuss why 100% efficiency is impossible.

Slide 9: STEM Challenge

12 min

Say: *"Design a power plan that keeps your phone alive for a full 8-hour school day. Every watt counts."*

Watch For

Groups plan schedules balancing usage needs with power conservation.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Screen Brightness, Processing Load
- **INTERNAL:** Battery Capacity, Heat Generation, Battery Life

Relationships

- **Screen Brightness to Battery Life:** NEGATIVE (-) — Higher screen brightness draws more power from the battery, converting more chemical energy to light (plus wasted heat), reducing the time the battery can last.
- **Processing Load to Heat Generation:** POSITIVE (+) — More intensive processing means more electrical energy flowing through transistors, and more energy lost as heat through electrical resistance. Gaming generates several watts of heat.
- **Heat Generation to Battery Life:** NEGATIVE (-) — Heat represents wasted energy — every watt of heat is a watt that didn't go to useful function. More heat means faster energy depletion and shorter battery life.

Key Discoveries

- Energy conversion from battery chemical energy to screen light and computation is never 100% efficient
- Heat generation is wasted energy — every watt of heat is a watt NOT powering your screen or apps
- Screen brightness and processing load have a multiplicative effect on battery drain
- Battery capacity degrades over time, meaning the same usage pattern drains faster on an old phone

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Closing apps saves battery"	On modern phones, background apps are 'frozen' and use almost zero power. Force-closing them actually WASTES energy because reopening them requires more processing than leaving them suspended. The real battery drains are screen brightness, cellular/GPS radios, and active processing.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Charging overnight damages the battery"	Modern phones have smart charging circuits that stop charging at 100% and maintain the battery. However, keeping a lithium-ion battery at 100% charge does cause slightly faster degradation over months. The bigger damage comes from extreme heat and fast-charging frequently.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Bigger battery always means longer life"	Battery size (mAh) is only half the equation. A phone with a 5,000 mAh battery but a power-hungry screen and processor might last shorter than a 3,000 mAh phone with efficient components. Total battery life = capacity ÷ average power draw. Efficiency matters as much as size.